Experiment 3

Aim : Apply Decision Tree and Random Forest for classification tasks

Dataset : https://www.kaggle.com/competitions/titanic/data?select=gender_submission.csv

Theory :

1. Decision Tree: The Logical Flowchart

Think of a **Decision Tree** as a game of "20 Questions." It looks at your data and finds the most important question to ask first (e.g., *"Was the passenger male or female?"*).

- **Splitting:** Based on the answer, it splits the data into two branches.
- **Leaves:** It keeps asking questions until it reaches a "leaf," which is the final prediction (Survived or Died).
- **The Visualization:** The tree diagram shows you the exact path a passenger takes. If a box is dark blue, the model is very confident they survived; if it's dark orange, it's confident they didn't.

2. Random Forest: Wisdom of the Crowd

A **Random Forest** is just a collection of many Decision Trees (a "forest") working together.

- **Diversity:** Each tree is trained on a slightly different random subset of the data.
- **Voting:** When it's time to make a prediction, every tree in the forest "votes." The forest picks the result that got the most votes.
- **Why it's better:** One tree might be "distracted" by a weird coincidence in the data (overfitting), but it's unlikely that 100 trees will all make the same mistake.

3. Understanding the Diagrams

Confusion Matrix

This is a "Scorecard." It doesn't just tell you the accuracy; it tells you **how** the model failed. It shows if the model is being too optimistic (predicting survival for everyone) or too pessimistic.

Feature Importance

This ranks your variables. It tells you which data points (like Sex or Ticket Price) actually helped the model make a decision and which ones were just "noise" (like Passenger ID).

ROC Curve

This measures "Separation Power." Imagine a fog lifting off a battlefield—the ROC curve shows how well the model can distinguish between the two groups as the "fog" clears. A perfect model would have a curve that goes straight to the top-left corner.

Code & Outputs :

Step 1 : Preprocessing

```
def preprocess(df):
    # Extract 'Title' (Mr, Mrs, Miss, etc.) from Name
    df['Title'] = df.Name.str.extract(' ([A-Za-z]+)\.', expand=False)
    df['Title'] = df['Title'].replace(['Lady', 'Countess','Capt', 'Col','Dr', 'Major', 'Rev', 'Sir'], 'Rare')
    df['Title'] = df['Title'].replace({'Mlle': 'Miss', 'Ms': 'Miss', 'Mme': 'Mrs'})

    # Fill Missing Data
    df['Age'] = df.groupby('Title')['Age'].transform(lambda x: x.fillna(x.median()))
    df['Embarked'] = df['Embarked'].fillna(df['Embarked'].mode()[0])
    df['Fare'] = df['Fare'].fillna(df['Fare'].median())

    # Mapping Categorical to Numerical
    df['Sex'] = df['Sex'].map({'female': 1, 'male': 0})
    df['Embarked'] = df['Embarked'].map({'S': 0, 'C': 1, 'Q': 2})
    df['Title'] = df['Title'].map({"Mr": 1, "Miss": 2, "Mrs": 3, "Master": 4, "Rare": 5}).fillna(0)

    return df[['Pclass', 'Sex', 'Age', 'SibSp', 'Parch', 'Fare', 'Embarked', 'Title']]
```

Step 2 : Data Preparation

```
X = preprocess(pd.read_csv('train.csv'))
y = pd.read_csv('train.csv')['Survived']
X_train, X_val, y_train, y_val = train_test_split(X, y, test_size=0.2, random_state=42)
```

Step 3 : Model Training

```
rf = RandomForestClassifier(n_estimators=100, max_depth=6, random_state=42)
rf.fit(X_train, y_train)
```

Step 4 : Evaluation

```
print(classification_report(y_val, rf.predict(X_val)))
```

print(f"ROC AUC Score: {roc_auc_score(y_val, rf.predict_proba(X_val)[:, 1]):.4f}")

Output :

```
--- Random Forest Performance ---
              precision    recall  f1-score   support

           0       0.83      0.89      0.86       105
           1       0.82      0.74      0.78        74

    accuracy                           0.83       179
   macro avg       0.83      0.81      0.82       179
weighted avg       0.83      0.83      0.83       179

ROC AUC Score: 0.9037
```

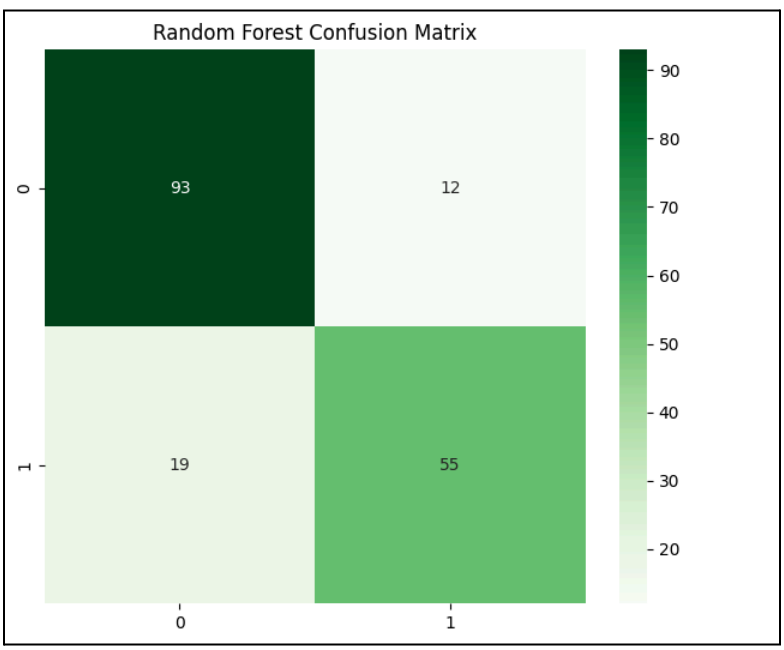
For Graphs :

Step 1. Confusion Matrix

```
plt.figure(figsize=(12, 5))
cm_rf = confusion_matrix(y_test, rf_model.predict(X_test))
sns.heatmap(cm_rf, annot=True, fmt='d', cmap='Greens')

plt.title('Random Forest Confusion Matrix')
plt.ylabel('Actual')
plt.xlabel('Predicted')
plt.show()
```

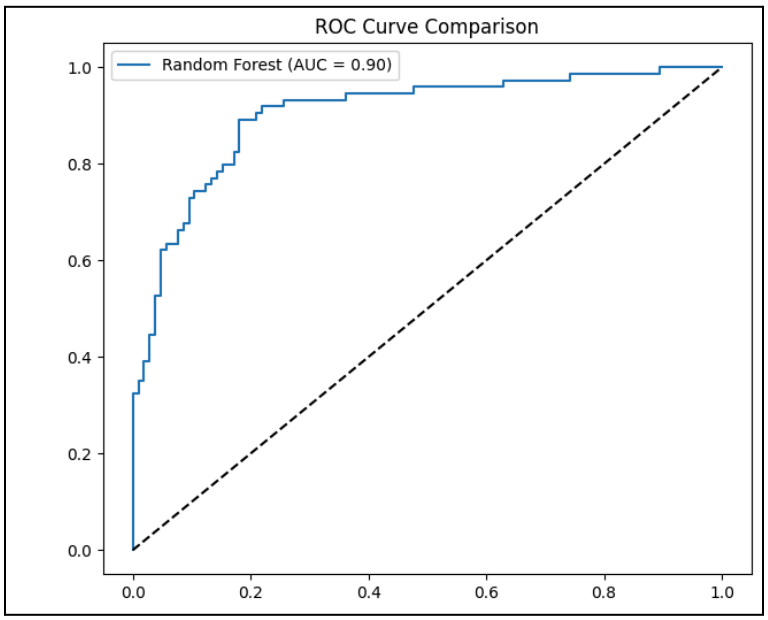
Output:



Step 2. ROC Curve

```
plt.figure(figsize=(8, 6))
probs = rf_model.predict_proba(X_test)[:, 1]
fpr, tpr, _ = roc_curve(y_test, probs)
plt.plot(fpr, tpr, label=f'Random Forest (AUC = {auc(fpr, tpr):.2f})')
plt.plot([0, 1], [0, 1], 'k--')
plt.title('ROC Curve Comparison')
plt.legend()
plt.show()
```

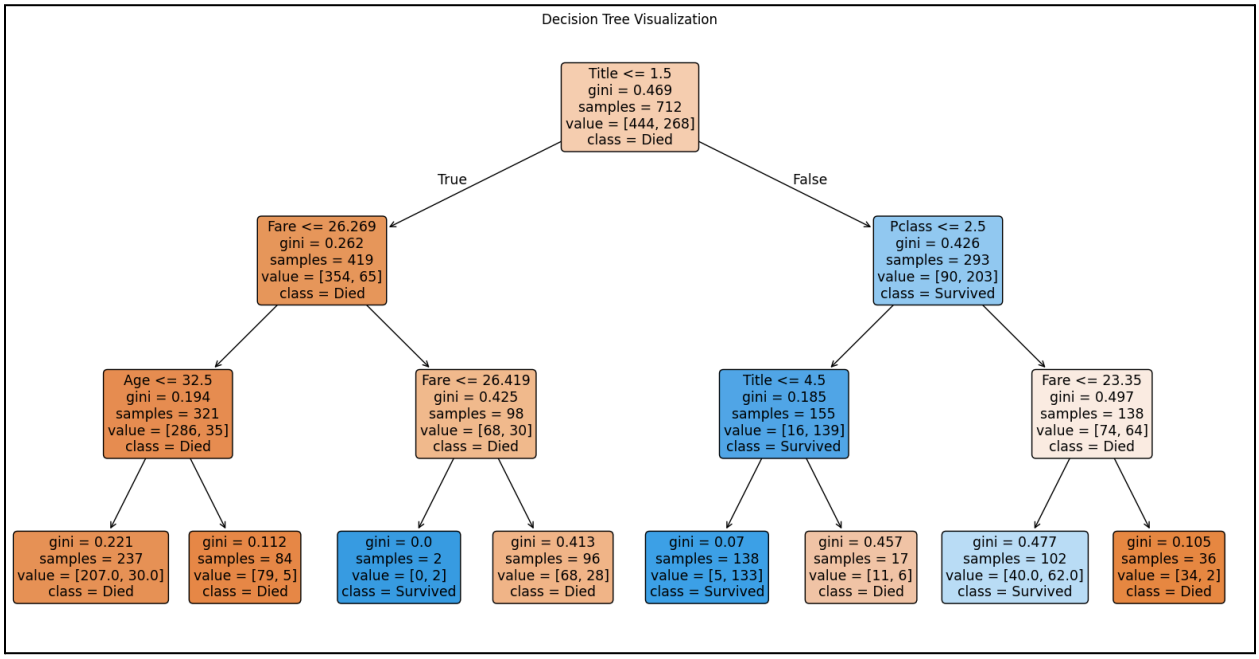
Output:



Step 3. Decision Tree Visualization

```
plt.figure(figsize=(20,10))
plot_tree(dt_model, feature_names=features, class_names=['Died', 'Survived'], filled=True)
plt.show()
```

Output :



Conclusion : Hence we have successfully completed the visualization of the decision tree vs the random forest algorithm